



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Fundamentals of Data Science

Course Code: DSA0404

Slot: A

Faculty: KRISHNARAJ M

Submitted By

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Hands on Data Summarization, Data Distribution, and Outlier Treatment.

1. Complete Summary of an Employee Dataset

Code:

```
employeeedata1.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/employeeedata1.py (3.12.3)
File Edit Format Run Options Window Help
import pandas as pd

data = {
    "Name": ["Arun", "Ravi", "Priya", "Kiran", "Anu"],
    "Age": [25, 30, 28, 35, None],
    "Salary": [30000, 45000, 40000, 55000, 35000],
    "Department": ["IT", "HR", "IT", "Finance", "HR"]
}

df = pd.DataFrame(data)

print("Employee Dataset:")
print(df)

print("\nNumber of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nStatistical Summary:")
print(df.describe(include="all"))
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/employeedata1.py
Employee Dataset:
   Name  Age  Salary Department
0  Arun  25.0   30000          IT
1  Ravi  30.0   45000          HR
2  Priya 28.0   40000          IT
3  Kiran 35.0   55000    Finance
4   Anu   NaN   35000          HR

Number of Rows: 5
Number of Columns: 4

Data Types:
Name          str
Age           float64
Salary        int64
Department    str
dtype: object

Missing Values:
Name          0
Age           1
Salary        0
Department    0
dtype: int64

Statistical Summary:
   Name  Age  Salary Department
count    5  4.000000    5.000000          5
unique    5    NaN          NaN          3
top  Arun    NaN          NaN          IT
freq     1    NaN          NaN          2
mean    NaN  29.500000  41000.000000    NaN
std     NaN   4.203173   9617.692031    NaN
min     NaN  25.000000  30000.000000    NaN
25%     NaN  27.250000  35000.000000    NaN
50%     NaN  29.000000  40000.000000    NaN
75%     NaN  31.250000  45000.000000    NaN
max     NaN  35.000000  55000.000000    NaN
>>>
```

2. Frequency Distribution of Students' Grades with Bar Chart

Code:

```
*FDofstudents.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDofstudents.py (3.12.3)*
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt

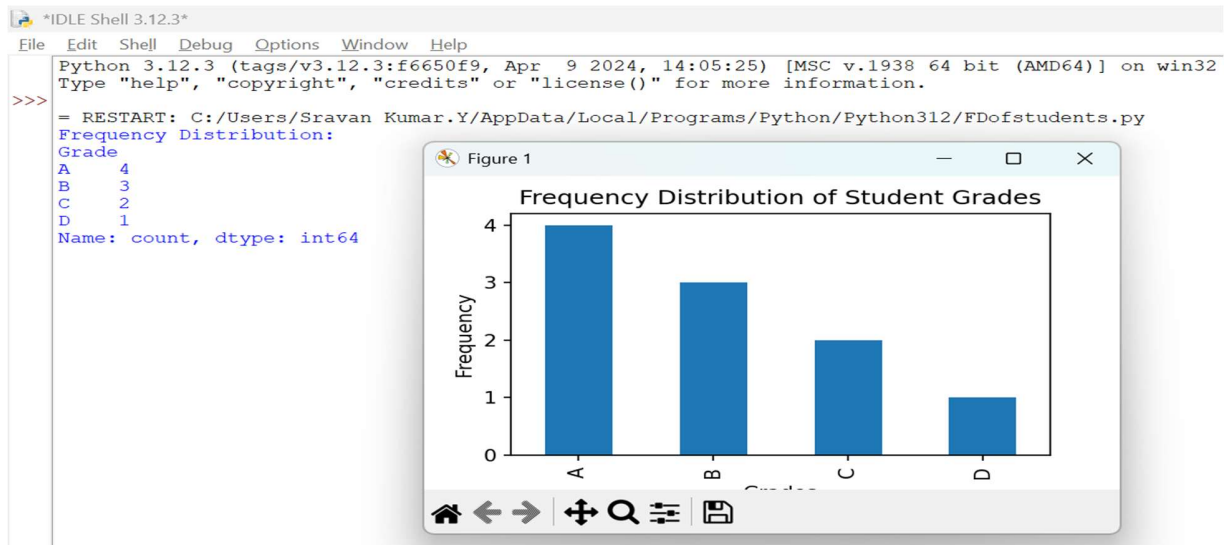
grades = ["A", "B", "A", "C", "B", "A", "B", "C", "A", "D"]
df = pd.DataFrame({"Grade": grades})
frequency = df["Grade"].value_counts().sort_index()

print("Frequency Distribution:")
print(frequency)

frequency.plot(kind="bar")

plt.title("Frequency Distribution of Student Grades")
plt.xlabel("Grades")
plt.ylabel("Frequency")
plt.show()
```

Output:



3. Histogram of House Prices and Identify Distribution

Code:

```
houseprices.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/houseprices.py (3.12.3)
File Edit Format Run Options Window Help

import pandas as pd
import matplotlib.pyplot as plt

prices = [
    20, 22, 23, 24, 25, 25, 26, 27,
    28, 30, 32, 35, 40, 50, 80
]

df = pd.DataFrame({"Price": prices})

plt.hist(df["Price"], bins=6, edgecolor="black")

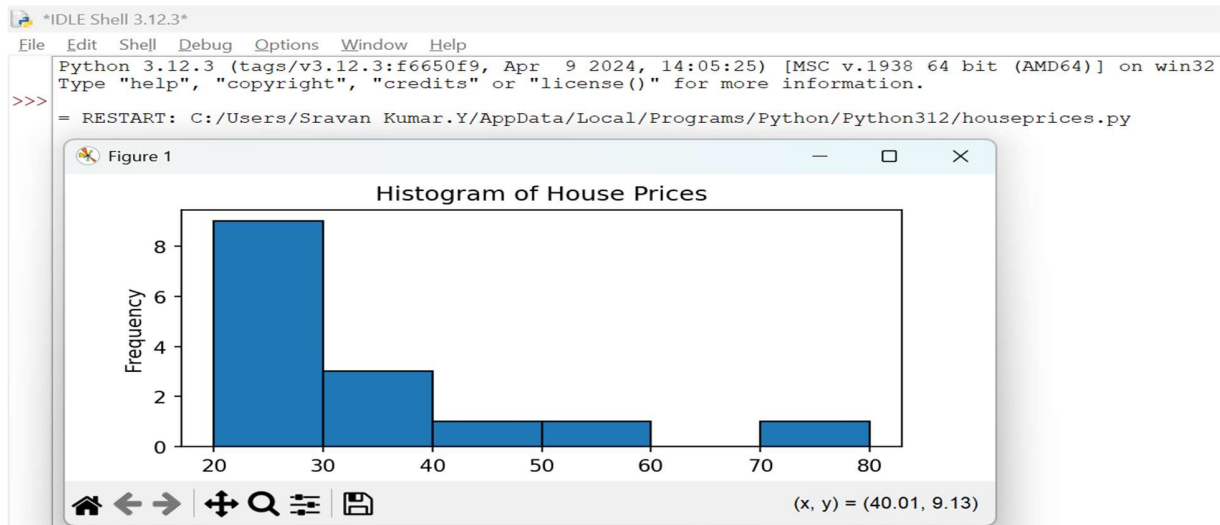
plt.title("Histogram of House Prices")
plt.xlabel("House Price")
plt.ylabel("Frequency")
plt.show()

skewness = df["Price"].skew()

print("Skewness:", skewness)

if skewness > 0.5:
    print("Data is positively (right) skewed.")
elif skewness < -0.5:
    print("Data is negatively (left) skewed.")
else:
    print("Data is approximately symmetric.")
```

Output:



4. Box Plot of Salary Data and Identify Outliers

Code:

```
salary.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/salary.py (3.12.3)
File Edit Format Run Options Window Help

import pandas as pd
import matplotlib.pyplot as plt

salary = [
    25000, 28000, 30000, 32000, 35000,
    36000, 38000, 40000, 42000, 100000
]

df = pd.DataFrame({"Salary": salary})

plt.boxplot(df["Salary"])

plt.title("Box Plot of Salary")
plt.ylabel("Salary")
plt.show()

Q1 = df["Salary"].quantile(0.25)
Q3 = df["Salary"].quantile(0.75)

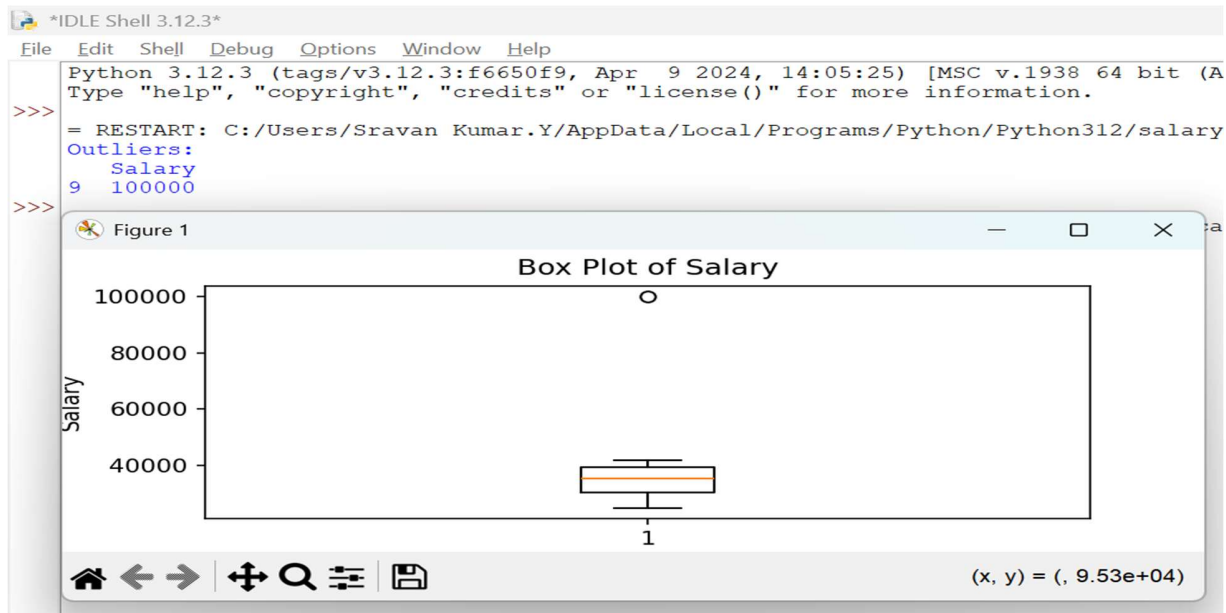
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[
    (df["Salary"] < lower) |
    (df["Salary"] > upper)
]

print("Outliers:")
print(outliers)
```

Output:



5. Detect Outliers in Students' Marks Using IQR

Code:

```
usingIQR.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/usingIQR.py (3.12.3)
File Edit Format Run Options Window Help
import pandas as pd

marks = [45, 50, 52, 55, 58, 60, 62, 65, 68, 70, 95, 10]

df = pd.DataFrame({"Marks": marks})

Q1 = df["Marks"].quantile(0.25)
Q3 = df["Marks"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[
    (df["Marks"] < lower) |
    (df["Marks"] > upper)
]

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)

print("\nOutliers:")
print(outliers)
```

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9 AMD64) on win32
Type "help", "copyright", "credits"

>>> = RESTART: C:/Users/Sravan Kumar.Y/
IQR.py
Q1: 51.5
Q3: 65.75
IQR: 14.25

Outliers:
Marks
10 95
11 10
>>>
```

6. Remove Outliers from a Sales Dataset Using IQR

Code:

```
removingoutliers.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/removingoutliers.py (3.12.3)
File Edit Format Run Options Window Help

import pandas as pd

sales = [100, 120, 125, 130, 140, 150, 160, 170, 180, 1000]
df = pd.DataFrame({"Sales": sales})

print("Original Dataset:")
print(df)

Q1 = df["Sales"].quantile(0.25)
Q3 = df["Sales"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

cleaned_df = df[
    (df["Sales"] >= lower) &
    (df["Sales"] <= upper)
]

print("\nCleaned Dataset:")
print(cleaned_df)

IDLE Shell 3.12.3
File Edit Shell Debug Optio
Python 3.12.3 (tags/
AMD64) on win32
Type "help", "copyri
>>>
= RESTART: C:/Users/
ingoutliers.py
Original Dataset:
Sales
0    100
1    120
2    125
3    130
4    140
5    150
6    160
7    170
8    180
9   1000

Cleaned Dataset:
Sales
0    100
1    120
2    125
3    130
4    140
5    150
6    160
7    170
8    180
>>>
```

7. Histograms and Box Plots Before and After Removing Outliers

Code:

```
boxplots.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312
File Edit Format Run Options Window Help

import pandas as pd
import matplotlib.pyplot as plt

data = [10, 12, 13, 14, 15, 16, 17, 18, 19, 20, 100]
df = pd.DataFrame({"Values": data})

# Before removing outliers
plt.hist(df["Values"], bins=6, edgecolor="black")
plt.title("Histogram Before Removing Outliers")
plt.show()

plt.boxplot(df["Values"])
plt.title("Box Plot Before Removing Outliers")
plt.show()

# Calculate IQR
Q1 = df["Values"].quantile(0.25)
Q3 = df["Values"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

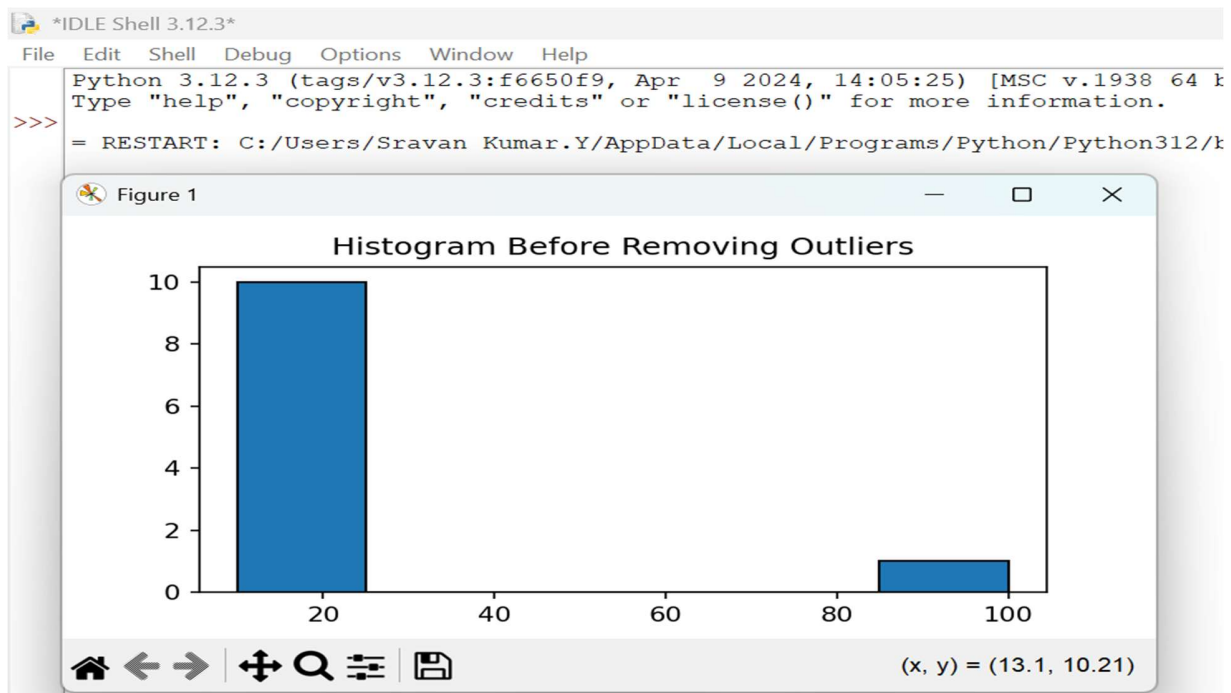
# Remove outliers
cleaned_df = df[
    (df["Values"] >= lower) &
    (df["Values"] <= upper)
]

# After removing outliers
plt.hist(cleaned_df["Values"], bins=6, edgecolor="black")
plt.title("Histogram After Removing Outliers")
plt.show()

plt.boxplot(cleaned_df["Values"])
plt.title("Box Plot After Removing Outliers")
plt.show()

print("Cleaned Dataset:")
print(cleaned_df)
```

Output:



8. EDA on Iris Dataset

Code:

eda.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/eda.py (3.12.3)

```
File Edit Format Run Options Window Help
plt.suptitle("Histograms of Iris Dataset")
plt.show()

# Box plots
df.boxplot()
plt.title("Box Plots of Iris Dataset")
plt.xticks(rotation=20)
plt.show()

# Detect and remove outliers using IQR
numeric_columns = iris.feature_names

cleaned_df = df.copy()

for column in numeric_columns:

    Q1 = cleaned_df[column].quantile(0.25)
    Q3 = cleaned_df[column].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = cleaned_df[
        (cleaned_df[column] < lower) |
        (cleaned_df[column] > upper)
    ]

    print("\nOutliers in", column)
    print(outliers)

    cleaned_df = cleaned_df[
        (cleaned_df[column] >= lower) &
        (cleaned_df[column] <= upper)
    ]

# Save cleaned dataset
cleaned_df.to_csv("cleaned_iris.csv", index=False)

print("\nOriginal Dataset Size:", df.shape)
print("Cleaned Dataset Size:", cleaned_df.shape)

print("\nCleaned dataset saved as cleaned_iris.csv")
```

Output:

